Analysis of a PLC-Based Conveyor System

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This project examines the use of programmable logic controllers (PLCs) in industrial automation through the design and simulation of a conveyor-based control system. The study focuses on how ladder logic is used to coordinate conveyor operation, sensor-based object detection, and stop/start control in a manufacturing-style environment. CODESYS was used to develop the PLC logic, while Factory I/O was used to simulate the automated process and visualize system behavior. The project analyzes the engineering principles behind discrete control, input/output logic, sequencing, and safe automation operation. A virtual conveyor application is presented as a case study to demonstrate how PLCs improve repeatability, efficiency, and reliability in industrial systems. The project also discusses practical limitations, including simulation-only testing, simplified plant dynamics, and the gap between virtual commissioning and physical implementation. This work shows how PLC-based automation remains a foundational technology in modern manufacturing and provides a useful framework for understanding larger industrial automation systems.

INTRODUCTION & MOTIVATION

Industrial automation plays a central role in modern manufacturing by improving productivity, repeatability, safety, and process consistency. Automated systems are widely used in conveyor lines, assembly stations, packaging systems, and material handling operations, where programmable logic controllers (PLCs) are commonly used to coordinate sensors, actuators, and machine sequences. Because of their reliability, structured logic, and widespread industrial adoption, PLCs remain one of the most important control technologies in automation engineering.

This project focuses on PLC-based conveyor automation as an example of industrial automation in a manufacturing-style environment. The system was developed using CODESYS for ladder logic programming and Factory I/O for virtual simulation of the conveyor process. The project examines how a PLC can be used to control conveyor motion, respond to sensor inputs, and implement start/stop logic in a predictable and organized way. This topic fits within the course area of Industrial Automation & Applications, particularly robotic or automated systems used in manufacturing and assembly lines.

The motivation for selecting this topic comes from both industry relevance and engineering value. In real manufacturing environments, automated conveyor systems are often part of larger robotic cells or production lines, where timing, sequencing, and safe operation are essential. Understanding how PLC logic is structured and applied helps connect classroom robotics and automation concepts to real industrial practice. This project is also meaningful because it goes beyond a high-level description and instead examines the engineering principles behind discrete control logic, sequencing, sensor-based decisions, and system behavior, which is exactly the type of technical depth expected in the project guidelines.

Another motivation for this project is that virtual commissioning tools such as Factory I/O allow automation systems to be tested and visualized before physical hardware is built. This makes simulation a useful learning tool for understanding how industrial systems behave, how control logic affects machine operation, and how design choices influence performance. By combining ladder logic development with simulated plant behavior, this project provides a practical and accessible way to study industrial automation while still emphasizing engineering analysis, limitations, and trade-offs.

BACKGROUND & THEORY

Industrial automation systems are designed to perform repetitive production tasks with high consistency, speed, and reliability. In manufacturing and assembly environments, automation often depends on control systems that coordinate machines, sensors, and actuators according to a defined logic sequence. This fits directly within the project category of Industrial Automation & Applications, especially robotic or automated systems used in manufacturing or assembly lines.

A central component in many industrial automation systems is the programmable logic controller (PLC). A PLC is an industrial digital controller used to monitor input signals, execute control logic, and update output devices in real time. Typical PLC inputs include pushbuttons, emergency stops, limit switches, and photoelectric or proximity sensors, while outputs may include motors, solenoids, relays, and indicator lights. In conveyor-based automation, the PLC acts as the decision-making unit that determines when the conveyor should run, stop, or respond to sensor-detected objects.

One of the most common PLC programming methods is ladder logic. Ladder logic is a graphical programming language based on relay-control concepts, where logic conditions are arranged in rungs that resemble an electrical ladder diagram. This format is widely used because it is easy to interpret, modular, and well suited for discrete-event control. In a basic conveyor application, ladder logic can be used to implement start/stop circuits, latching behavior, interlocks, sensor-triggered actions, and emergency-stop conditions.

The type of control used in this project is mainly discrete sequential control. Unlike continuous control systems, which regulate variables such as speed, temperature, or position through continuous mathematical feedback, discrete control systems operate based on logic states such as ON/OFF, TRUE/FALSE, or open/closed. Conveyor automation is a good example because machine actions are triggered by specific events, such as pressing a start button or detecting an object at a sensor. The PLC repeatedly scans the system by reading inputs, solving the logic program, and updating outputs. This scan cycle is what allows the automated process to behave predictably and repeatedly.

Another important concept is the role of sensors in automation. Sensors provide the PLC with information about the state of the system. In a conveyor application, a photoelectric or reflective sensor may detect the presence of a box, causing the controller to stop the belt, count an item, or trigger the next sequence in the process. Sensor-based logic is essential in industrial systems because it allows automation to respond dynamically to changing conditions rather than simply running on fixed timing alone.

This project also uses virtual commissioning, meaning the automation logic is developed and tested in simulation before physical hardware is built. The course project specifically allows the use of diagrams, simulations, and flowcharts where appropriate, and expects technical models rather than only high-level description. In this work, CODESYS was used to create and test the ladder logic, while Factory I/O was used to simulate the conveyor system and visualize system behavior. This approach is valuable because it allows the designer to verify logic sequences, test sensor interactions, and identify control issues before physical implementation. It also reflects modern industrial practice, where simulation is often used to reduce commissioning time and improve system understanding.

From an engineering standpoint, the most important theoretical ideas in this project are not just that the conveyor moves, but how and why the control system works. The project guidelines specifically emphasize explaining engineering principles, including models or diagrams where relevant, and discussing limitations, assumptions, and trade-offs. For that reason, this report focuses not only on describing ladder logic and PLC programming, but also on showing how sensor logic, latching, sequencing, and simulation-based validation contribute to practical industrial automation.

SYSTEM AND METHOD OF ANALYSIS

This project analyzes a PLC-based conveyor automation system developed in CODESYS and tested in Factory I/O. The purpose of the system is to demonstrate how programmable logic control can be used to automate a simple industrial material-handling process using ladder logic, input/output mapping, and event-based sequencing. This section focuses on the engineering method used to create the system and explains how the control logic operates, rather than only describing what the conveyor does.

The overall system can be described as a closed automation loop made up of four main parts: operator inputs, PLC logic, simulated plant behavior, and sensor feedback. Operator commands such as start, stop, and emergency stop are treated as PLC inputs. These inputs are read by the PLC program in CODESYS, which executes ladder logic instructions and determines whether the conveyor motor output should be energized. The output state is then sent to the Factory I/O model, where the simulated conveyor responds by running or stopping. A sensor placed along the conveyor provides feedback by detecting the presence of a box, allowing the logic to change the conveyor state based on the position of the object. The interaction between the controller, plant, and feedback signals can be summarized using the overall system block diagram shown in Figure 1.

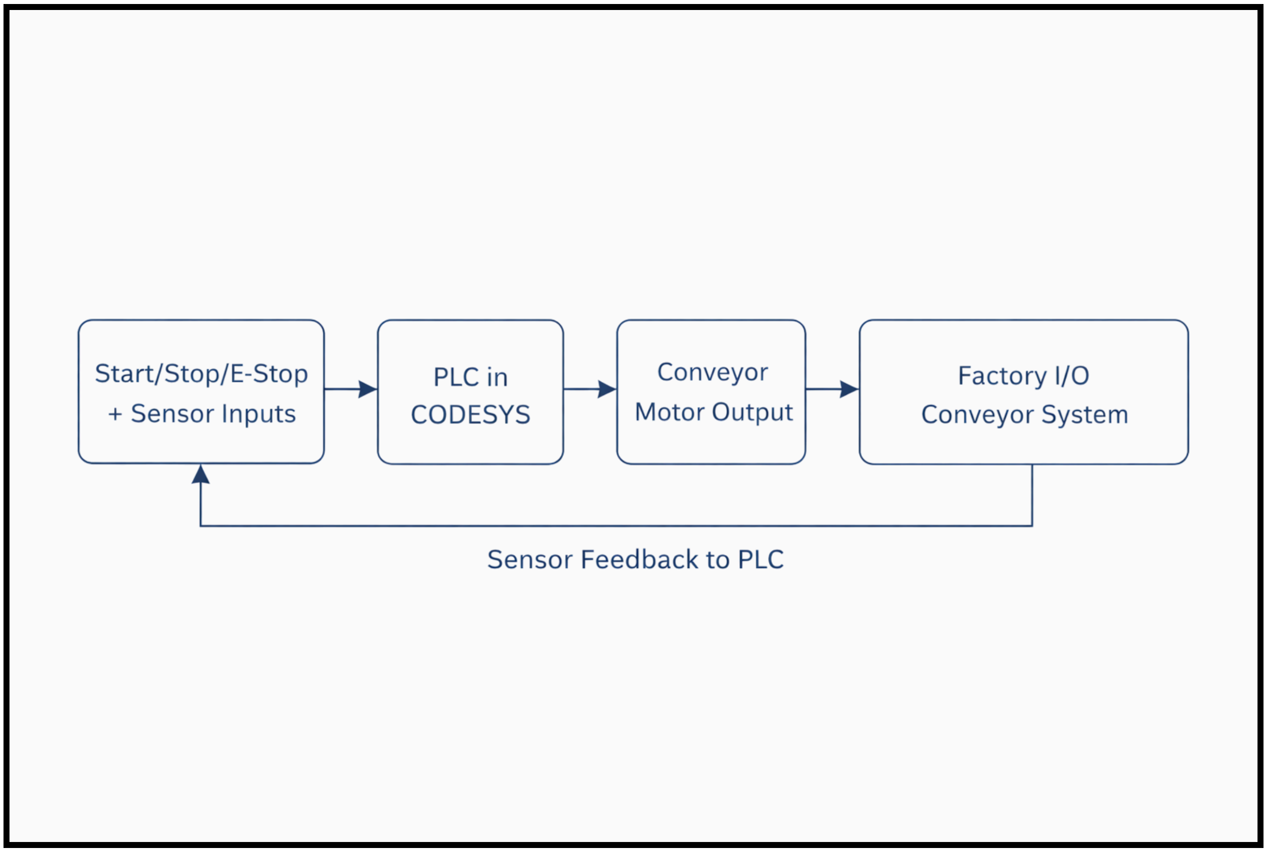

Figure 1. Block diagram of the PLC-based conveyor automation system showing operator inputs, PLC execution in CODESYS, conveyor output, Factory I/O plant behavior, and sensor feedback.

Table 1. Main Inputs and Outputs Used in the PLC-Based Conveyor System

Signal Name	Type	Role in the System
Start Button	Digital input	Starts conveyor operation

Stop Button	Digital input	Stops conveyor operation
Emergency Stop	Digital input	Forces conveyor shutdown and prevents unsafe motion
Reflective Sensor	Digital input	Detects box position for sensor-based control
Entry Sensor	Digital input	Detects parts entering the conveyor section
Exit Sensor	Digital input	Detects parts leaving the conveyor section
Conveyor Motor Output	Digital output	Drives the main conveyor
Buffer Conveyor Output	Digital output	Controls conveyor behavior in the buffer section
Start Light	Digital output	Indicates system start/run condition

The control strategy used in this project is based on discrete-event logic. In this type of system, machine actions are triggered by logical conditions rather than continuous mathematical control of variables such as speed or position. For example, if the start pushbutton is pressed and the emergency stop condition is not active, the conveyor is allowed to run. If the stop button is pressed, the run condition is removed. If the emergency stop is activated, the conveyor is immediately disabled and prevented from restarting until reset. This type of logic is common in industrial automation because it is reliable, easy to troubleshoot, and well suited for repetitive manufacturing tasks. To illustrate the core control idea, Figure 2 shows a simplified ladder rung used to energize the conveyor output based on operator input conditions. In this basic form, the conveyor is allowed to run only when the required logical conditions are satisfied, demonstrating the discrete ON/OFF nature of PLC-based control.

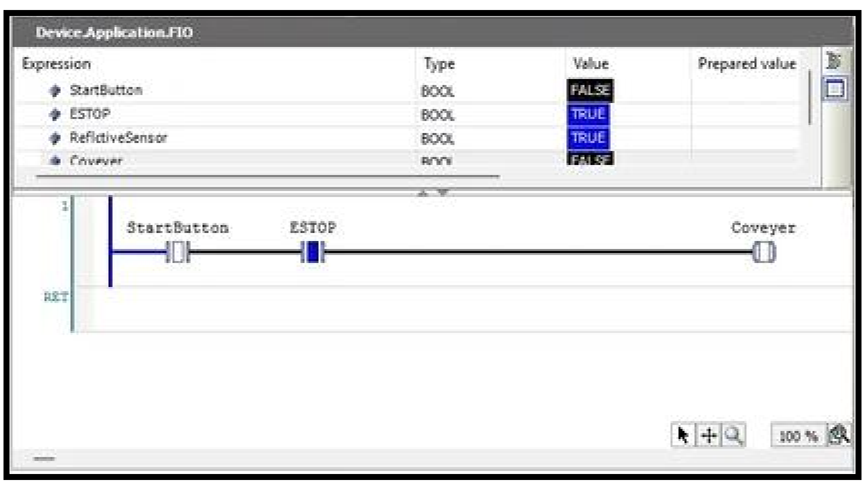

Figure 2. Basic ladder logic rung used for conveyor control in CODESYS, showing start-button input, emergency-stop interlock, and conveyor output.

A major part of the method is the use of a latching rung in ladder logic. The latch allows the conveyor to continue running after the momentary start pushbutton is released. Once the motor output is energized, an auxiliary logic condition maintains the run state until a stop condition breaks the rung. This is one of the most fundamental control patterns in industrial PLC programming because it allows a system to maintain operation without requiring a human operator to hold the start button continuously. In engineering terms, the latch creates a memory function within the discrete control sequence.

The project also includes sensor-based automation behavior. In the automated version of the system, a reflective or presence-type sensor is used to detect a box moving on the conveyor. The PLC uses this sensor signal as part of the control logic so that the conveyor can stop automatically when the object reaches a target position. This demonstrates how industrial automation systems combine actuator commands with sensor feedback to create more intelligent machine behavior. Instead of relying only on fixed timing, the system responds to the actual state of the process. This improves repeatability and better reflects how real conveyor systems operate in industrial environments. In Figure 3, the seal-in branch provides memory for the run condition, while the emergency-stop and sensor inputs act as additional logical conditions that influence conveyor operation.

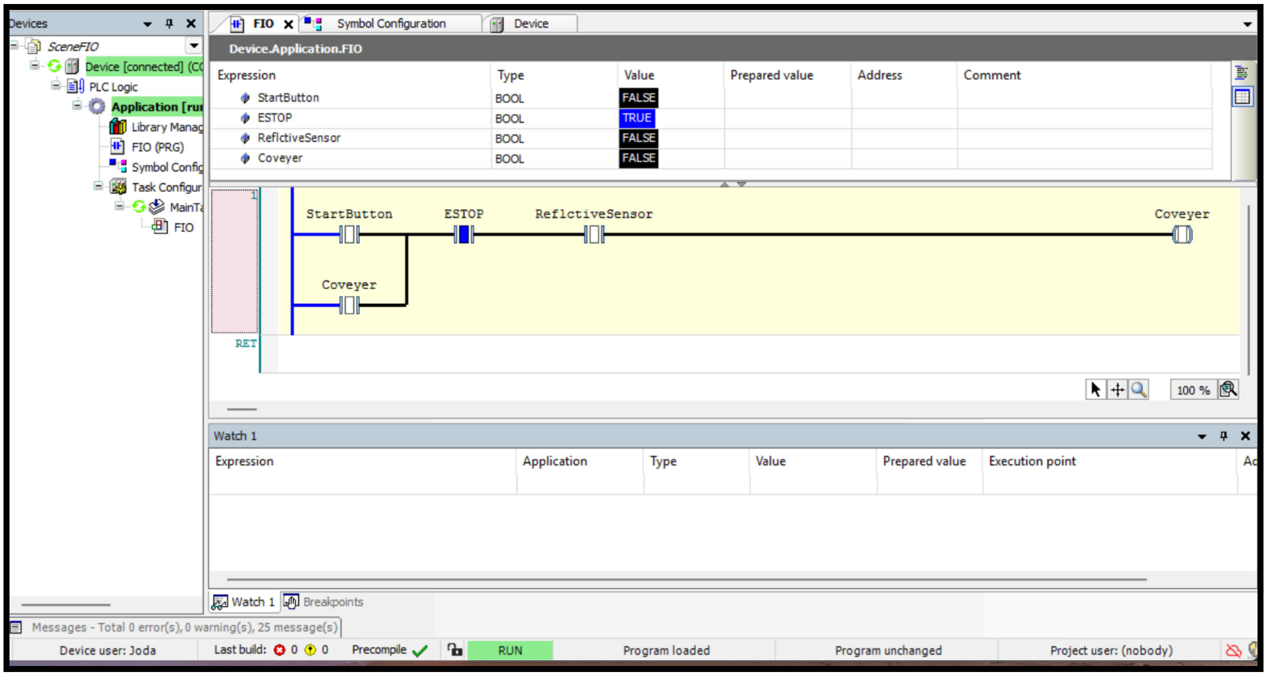

Figure 3. Main conveyor control rung in CODESYS including the latching branch, emergency-stop condition, reflective sensor input, and conveyor output.

Another important aspect of the method is the use of virtual commissioning. Rather than building a physical conveyor station, the system was implemented and validated through simulation. This approach is acceptable for the project because the assignment specifically expects technical models, diagrams, simulations, or flowcharts where appropriate, and it allows students to focus on engineering reasoning without requiring full hardware implementation. By using CODESYS together with Factory I/O, the project simulates the interaction between controller logic and machine behavior in a realistic way. This makes it possible to test ladder logic, verify I/O responses, and observe how design changes affect system operation before physical hardware is involved.

From a system-analysis perspective, the conveyor control method involves several engineering considerations. First, the logic must be structured so that unsafe or unintended motion does not occur during stop or emergency-stop conditions. Second, the I/O mapping between the PLC program and the simulation must be correct, since an input or output mismatch could cause the wrong machine response. Third, the control sequence must be simple enough to remain readable and maintainable, which is one reason ladder logic remains widely used in industry. Finally, the method must be evaluated not only in terms of whether it works, but also in terms of its assumptions and limitations. The project guidelines specifically emphasize discussing limitations, assumptions, and trade-offs as part of the analysis.

This system fits within the course topic of Industrial Automation & Applications, particularly robotic automation in manufacturing or assembly-line style environments. Although the project is based on a conveyor rather than a full industrial robot manipulator, it still represents a core automation problem: coordinating inputs, logic, outputs, and process flow in a way that is structured, repeatable, and safe. In that sense, the conveyor serves as a practical case study for larger automated production systems.

APPLICATION EXAMPLE

The application example used in this project is a simulated conveyor automation system representing a basic industrial material-handling process. Conveyor systems are widely used in manufacturing, packaging, sorting, and assembly lines because they provide a simple and reliable method for moving parts or products between workstations. This makes them a strong example of industrial automation in manufacturing or assembly lines, which is one of the suggested project areas in the assignment.

In this case study, the conveyor system was developed using CODESYS for the control logic and Factory I/O for the simulation environment. The purpose of the example was to show how a PLC can coordinate system behavior through ladder logic using operator commands and sensor inputs. Rather than focusing on a complex production system, the project uses a simpler conveyor process to clearly demonstrate key automation principles such as latching, interlocks, sensor-based stopping, and emergency-stop behavior. Figure 4 shows how the PLC logic on the left is linked to the simulated conveyor system on the right, allowing the effect of ladder logic decisions to be observed directly in the virtual industrial environment.

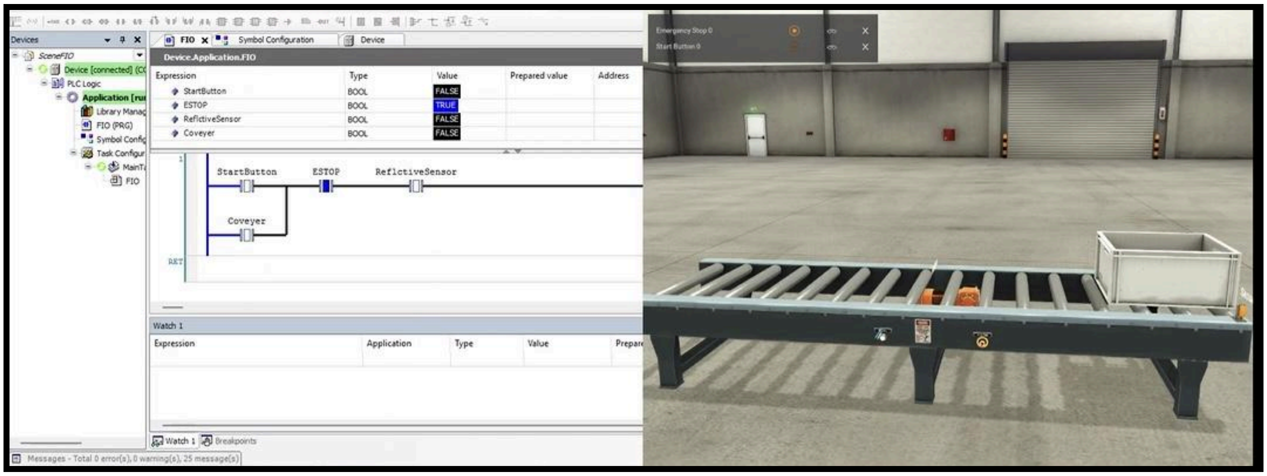

Figure 4. Combined view of the CODESYS ladder logic environment and the Factory I/O conveyor simulation used in the project case study.

The first operating mode in the simulation is a manual conveyor control mode. In this version, the system uses a start button, stop button, and emergency stop input. When the start pushbutton is pressed, the ladder logic energizes the conveyor motor and a latch keeps the conveyor running after the pushbutton is released. Pressing the stop button removes the motor run condition and halts conveyor motion. If the emergency stop is activated, the logic immediately disables the conveyor and prevents restart until the system is reset. This operating mode demonstrates the basic structure of an industrial PLC control circuit and shows how discrete logic is used to create predictable machine behavior.

The second operating mode adds automatic sensor-based control. In this version, a box travels along the simulated conveyor until it reaches a reflective or presence-type sensor. Once the sensor detects the box, the PLC logic changes the output condition and stops the conveyor automatically. This mode is important because it demonstrates a transition from pure operator-driven control to feedback-based automation. In real industrial systems, this type of logic is used for part positioning, counting, sorting, buffering, and transfer operations. The case study therefore shows how even a simple conveyor can represent a broader automation problem involving machine sequencing and event-driven control.

A third mode of the project extended the conveyor example beyond simple start/stop and sensor-based stopping by incorporating part-counting and buffer logic. In this version, counter instructions were used to track items detected at different positions in the system, such as entry and exit points. Comparison logic was then used to determine whether a conveyor section should continue running based on the number of parts currently on the belt or stored in the buffer area. This made the case study more representative of a real industrial automation task, since many conveyor systems do not only move objects, but also manage part flow, temporary accumulation, and controlled transfer between zones. By adding counters and logical comparison blocks, the project demonstrated a more advanced form of sequencing and process control than a basic single-rung conveyor example. Figure 5 shows the more advanced ladder implementation used in the third operating mode of the project, where counters and comparison logic were added to manage part flow and buffered conveyor behavior.

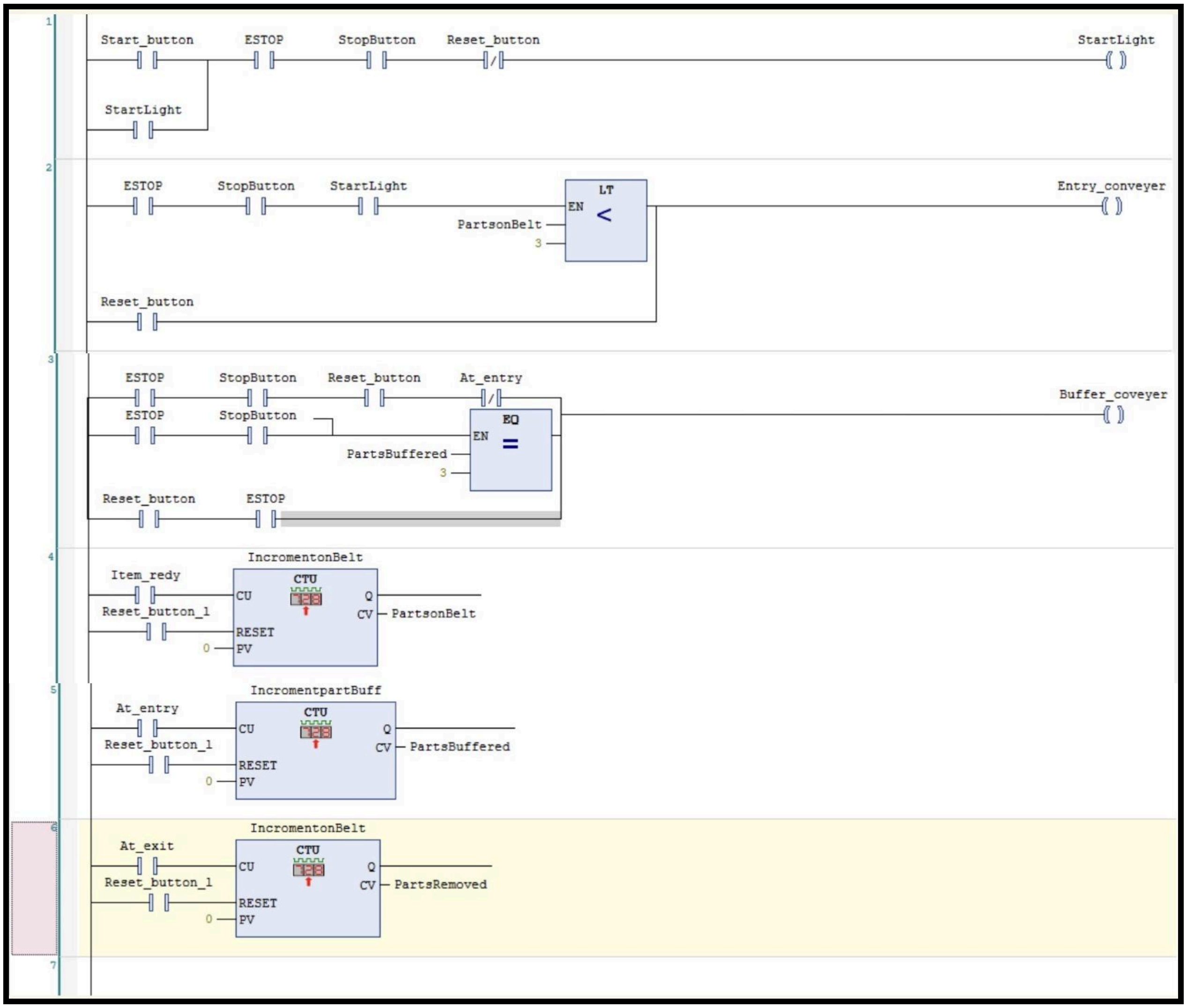

Figure 5. Expanded ladder logic used for the conveyor case study, including start/stop control, counter instructions, comparison logic, and buffered part-handling behavior.

A useful engineering feature of this example is that it allows direct observation of the relationship between input conditions, PLC logic, and machine response. In the simulation, the user can see that each control decision corresponds to a change in ladder logic state and to a visible physical action in the conveyor model. This is one reason the project works well as a teaching and analysis tool. The simulation makes it easier to verify whether the logic is functioning correctly and to understand how industrial control decisions affect a process in real time.

This case study also demonstrates several practical engineering decisions. First, the logic started relatively simple and expanded with complexity so that system operation would remain readable and easy to explain. Second, the emergency stop was treated as a high-priority condition to reflect safe automation practice. Third, the sensor was used as a condition for automatic stopping so that the process would respond to the actual location of the box rather than depending only on timing assumptions. These choices improve clarity, repeatability, and relevance to real industrial automation.

The system captures key features of larger automated systems. Industrial automation often begins with the same building blocks shown here: pushbuttons, sensors, motor outputs, interlocks, sequencing, and controller logic. More advanced systems may add robot arms, vision systems, variable-speed drives, or networked communication, but they are still built on the same core principle of coordinating inputs and outputs through structured control logic. For that reason, the conveyor simulation provides a realistic and useful case study for understanding the foundations of PLC-based automation.

This application example also supports the presentation component of the project. The video demonstration helps show the actual operation of the system and strengthens the connection between theory, ladder logic, and simulated industrial behavior.

DISCUSSION

The conveyor project worked well as a practical example of PLC-based industrial automation. The different operating modes showed a clear progression in complexity, starting with basic start/stop and emergency-stop control, then moving into sensor-based stopping, and finally into a more advanced mode that used counters and comparison logic to manage part flow. Because of that progression, the project was useful not only for demonstrating that the conveyor could run, but also for showing how ladder logic can be expanded step by step into a more structured automation sequence.

One of the strengths of the project was that the logic remained understandable while still demonstrating several important control ideas. The basic start/stop arrangement showed how a PLC can respond to operator inputs in a predictable way. The latching branch showed how a momentary pushbutton can be used to maintain conveyor operation without requiring continuous input. The emergency stop condition added an important safety-related control layer by forcing the conveyor to shut down when required. These are simple ideas, but they are also foundational in real automation systems.

The sensor-based mode added another level of control by allowing the conveyor to respond to the actual position of a box rather than only to operator commands. That made the system more realistic, since industrial automation often depends on feedback from the process itself. Instead of relying only on fixed timing, the conveyor could stop when a sensor detected that the part had reached a certain location. This kind of event-based response is important in material handling applications because it improves repeatability and allows machine actions to depend on real process conditions.

The third mode increased the technical depth of the project by introducing counters and logic for part handling. In this version, the system did more than simply move a box from one point to another. It also tracked parts and used comparison logic to decide how conveyor sections should behave based on how many items were present in different parts of the system. That made the project feel closer to an actual industrial automation problem, where the controller often has to manage flow, accumulation, and movement between zones rather than just turn a motor on or off. This part of the project showed that ladder logic can scale from very simple machine control to more organized process sequencing.

Another strength of the project was the use of simulation. CODESYS and Factory I/O made it possible to test the logic, observe system response, and adjust the program without needing to build a physical conveyor setup. That was useful because it allowed the control behavior to be checked visually and made the relationship between the PLC logic and the conveyor motion easier to understand. It also made the project easier to explain, since the software showed both the ladder logic and the machine response clearly.

At the same time, the project has limitations. The most obvious one is that the system was only tested in simulation. Even though the logic performed as expected in CODESYS and Factory I/O, a simulated model does not include all the issues that appear in real hardware. In a physical system, there could be wiring mistakes, sensor alignment problems, electrical noise, motor delays, mechanical wear, or timing differences that do not appear as strongly in a virtual environment. Because of that, the results should be viewed as a strong simulation-based demonstration, but not as final proof that the same system would work perfectly on a real conveyor without additional testing.

Another limitation is that the project still represents a simplified automation system. Even with the added counter logic, it does not include many features found in larger industrial systems, such as variable-speed drives, fault monitoring, alarm handling, communication over industrial networks, or integration with robot arms and higher-level control systems. Adding those features would make the system more realistic, but it would also make it much more complicated. Keeping the project focused on ladder logic, conveyor control, sensor input, and part counting was a reasonable trade-off because it kept the main control ideas visible and easier to analyze.

There are also assumptions built into the system. The project assumes that the sensors detect parts correctly, that outputs respond as expected, and that the simulated conveyor behaves consistently from cycle to cycle. In a real plant, those assumptions are not always true. Sensors can miss parts or trigger falsely, motors may not respond instantly, and physical systems can behave differently depending on load, wear, or operating conditions. This matters because automation design depends on more than just correct logic on the screen. It also depends on how reliably the physical process responds in practice.

One challenge in the project was making sure the PLC logic matched the simulation setup correctly. Input and output mapping had to be assigned properly or the system would not behave as intended, even if the ladder logic itself looked correct. Another challenge was keeping the program readable while still adding more functionality. As the project moved from basic control into sensor response and then into counters and comparison logic, the logic naturally became more involved. That made it more important to organize the ladder rungs clearly so the system could still be understood and explained. And finally, The most difficult part of the project by far was getting CODESYS and Factory I/O configured and communicating correctly. The setup process was more involved than I initially expected and required configuring the software, licensing, device setup, and communication structure in the correct sequence so both environments could exchange data properly. Once that connection was established, the rest of the project became much smoother.

Overall, the project showed that even a relatively small conveyor example can demonstrate several important automation concepts. It covered basic operator control, safety-related shutdown behavior, sensor-based feedback, and more advanced part-counting logic within one system. While it is still much simpler than a full industrial production line, it does show how PLC ladder logic can be used to build automation in a structured and expandable way. That is probably the most useful takeaway from the project: the same control framework can begin with very simple logic and then grow into more capable automation behavior as new functions are added.

CONCLUSION

This project examined PLC-based conveyor automation through the design and simulation of a control system developed in CODESYS and tested in Factory I/O. The project showed how ladder logic can be used to build conveyor control in a structured way, beginning with basic start/stop and emergency-stop behavior, then expanding into sensor-based stopping and more advanced counter-based logic for part handling. By working through these different stages of control, the project demonstrated how a PLC can be used not only to turn a conveyor on and off, but also to manage process flow in a more organized and repeatable way.

One of the main takeaways from the project is that even a relatively small conveyor example can illustrate several important industrial automation concepts. The system demonstrated discrete-event control, latching logic, interlocks, sensor feedback, and counting logic within one application. It also showed that good automation design is not just about making a system operate, but about developing control logic that is understandable, maintainable, and scalable as more functions are added. In that sense, the project moved beyond a simple demonstration and became a useful case study in how ladder logic can grow from basic machine control into more structured automation behavior.

The project also showed the value of simulation as both a design and learning tool. Using CODESYS together with Factory I/O made it possible to test the ladder logic, observe the effect of control decisions, and study the relationship between the PLC program and the conveyor system without building a full physical setup. At the same time, the project made it clear that simulation has limits. Real systems involve issues such as wiring errors, sensor reliability, timing variation, and mechanical behavior that are not always captured fully in a virtual environment.

Overall, the project showed that PLC-based automation remains a practical and important part of industrial control. Although the system studied here was smaller and simpler than a full production line, it still captured many of the same core ideas used in larger automation systems: operator control, safety-related shutdown logic, process feedback, sequencing, and part tracking. For that reason, the conveyor case study provided a useful way to study industrial automation while also showing how ladder logic can be expanded step by step into more capable control behavior.

REFERENCES

1. International Electrotechnical Commission. (2025). IEC 61131-3:2025 Programmable controllers—Part 3: Programming languages. IEC.
2. Petruzella, F. D. (5th ed.). Programmable Logic Controllers. McGraw-Hill.
3. Real Games. (n.d.). Factory I/O manual. Factory I/O Documentation.
4. Real Games. (n.d.). Getting started. Factory I/O Documentation.
5. CODESYS Group. (n.d.). CODESYS development system documentation. CODESYS Documentation.
6. ABB. (n.d.). Overview of the IEC 61131 standard.